BDS23124

DATA MINING AND DATA WAREHOUSING

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PROJECT

TELECOM SERVICE PROVIDER CHURN IN MALAYSIA

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Table of Contents

Chapter 1: Introduction 3
1.1 Problem Statement (scope validation) 3
Chapter 2: Databases Description 5
2.1 Data Source and Implementation 5
2.2 Feature Description of the Datasets 6
2.3 Target Variable of the Dataset 8
2.4 Exploratory Data Analysis (EDA) 9
2.4.1 EDA on Customer Master Table 9
2.4.2 EDA on Usage Transaction Table 12
2.4.3 EDA on Churn Decision Table 15
Chapter 3: Implementation and Evaluation 17
3.1 Processing Workflow of Orange Tool 17
3.2 Models Implementations 18
3.3 Evaluation Metrics of Results 18
Chapter 4: Performance Analysis and Discussion 20
4.1 Suggestion after Data Mining 20
4.2 Conclusion 20
Chapter 5: System Design and Prototype 22
5.1 Scope of System Design 22
5.2 System Architecture Overview 22
5.3 System Components 23
5.4 Key Features 25
5.5 Conclusion 27
Chapter 6: References 28

Chapter 1: Introduction

The telecommunications sector in Malaysia is very competitive and customers are likely to be able to move between mobile network operators (MNOs) including CelcomDigi, Maxis, U Mobile, Unifi (TM), and YES because of the similar services they offer, intensive marketing activities, and low switching costs.

Such a phenomenon is referred to as customer churn, which is a very serious problem to service providers since acquiring a new customer is much more expensive than keeping one.

Although highly developed 4G and 5G networks are widely adopted at the national level as part of the JENDELA effort, churn is a persistent problem caused by price dissatisfaction, network coverage, and customer service quality. Malaysian telecom operators are increasingly under pressure to retain subscribers, maximize service quality, and minimize loss of revenue due to churn.

A large part of available churn data, however, is global or synthetic in nature, and may not be applicable to Malaysia and its realities in terms of demographics, geography, as well as provider-specific issues. As such, this project is aimed at building a credible Malaysia-contextualized data that depicts the actual user trends, distribution of providers in the region and actual churn rates.

The objective is to use data-mining methods to determine the most effective variables contributing to churn and create a predictive model that can be used to predict customer defection. This will base the data-driven retention strategy and decision support in the field of Malaysian telecommunications.

1.1 Problem Statement (scope validation)

To design and construct a reliable, Malaysia-contextualized telecom churn dataset and to develop predictive data-mining models capable of identifying customers who are likely to switch telecom providers.

Area	Description
Domain	Malaysian Telecommunication Industry
Geographical Focus	Malaysia, covering major service providers such as Celcom Digi, Maxis, U Mobile, Unifi (TM), and YES.
Data Sources	Combination of open datasets (e.g., *Kaggle's Telco Customer Churn* dataset) and public Malaysian statistics (e.g., *Department of Statistics Malaysia (DOSM)*, *MCMC Telecommunications Statistics*).
Database Components	Three relational tables: Customer Master, Usage Transaction, and Churn Decision

Data Mining Models	Decision Tree, Random Forest, and Regression Analysis
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC/AUC
Tools / Environment	Orange 3 (visual analytics), Python (scikit-learn)
Deliverables	Realistic telecom churn dataset for Malaysia, Predictive model analysis and performance comparison, visualizations and dashboards, Prototype decision-support system, and Final report and documentation.

Table 1 Scope of Project

Chapter 2: Databases Description

2.1 Data Source and Implementation

The IBM Telco Customer Churn data comprises approximately 7,000 customer records of a California telecommunications company, which include customer demographics, location, services rendered, billing details, customer satisfaction ratings, and customer churning or not. The reason behind the dataset is to know why customers abandon the service. Important trends indicate that customers whose contracts are on a monthly basis, those with low scores of satisfaction, and those with high service fees are more likely to churn. The data is tidy, informative and can be used in the process of data mining like classification and churn prediction. (Team, 2019)

The DOSM dataset named as 'Population Malaysia' offers the official proportions of population per state of the country. The sampling distribution we used in our telco customer generator script was shaped using this dataset, on the attribute of state. The states of every customer were randomly selected not by chance, but based on the percentages of the real population provided by DOSM which makes the synthetic dataset closer to a more realistic representation and more representative. By matching the state distribution to the official demographic statistics, the resulting telco churn dataset will be more representative of the actual conditions in the Malaysian market, and the further data mining and churn analysis will be more valuable and plausible. (Data Catalogue | OpenDOSM, 2024)

Besides, The Telecommunications Statistics Dataset by DOSM providing an official overview of the national telecom landscape and long-term industry trends. Although the dataset does not contain individual customer-level churn information, it offers valuable context such as the number of broadband subscriptions, mobile usage levels, and the decline of traditional services like public payphones. These trends help explain how the Malaysian telecom market has evolved and why customers may be switching providers. For example, rapid growth in mobile broadband and fibre services suggests increased competition, which can influence churn behaviour as users move to providers offering faster speeds or better packages. By understanding these national-level patterns, my churn analysis becomes

more meaningful because it links customer churn to broader industry developments and competitive pressures within Malaysia's telecom sector. (Telecommunications Statistics, Malaysia - MAMPU, 2021)

2.2 Feature Description of the Datasets

Dataset design (schema)

Column	Description
customer_id (PK)	CUST100000…
age	int, 18–70
gender	Male/Female
State	Johor, Selangor, KL, Penang, Sabah, Sarawak
Provider	CelcomDigi, Maxis, U Mobile, Unifi (TM), YES
plan_type	Prepaid, Postpaid, Broadband, Family
contract_type	Month-to-month, 12-month, 24-month
internet_service	(4G, 5G, Fibre) — biased to urban states for 5G; Fibre for Broadband/Family
tenure_months	0–120; exponential skew to short tenures
monthly_fee_rm	RM, realistic by plan & provider

Table 2 customer_master_my_telco.csv

Column	Meaning / Description
customer_id (FK)	CUST100000…
data_usage_gb	log-normal; higher for 5G/Fibre; Family > Prepaid
calls_made, sms_sent	Poisson, modest
bill_amount_rm	≈ monthly_fee ± small noise
payment_method	Online Banking, Auto Debit, Credit Card, Cash
late_payment_count	0–4; higher for Cash/Online Banking, short tenures, higher bills
customer_service_calls	Poisson low
complaint_count	0–3; binomial
network_quality_score	1–5; slightly higher in KL/Selangor/Penang/Johor, varies by provider

Table 3 usage_transaction_my_telco.csv

Column	Meaning / Description
customer_id (FK)	CUST100000…
Churn	(Yes/No) — base ~20% adjusted by risk drivers below
satisfaction_score	1–10; inverse of churn risk + noise

reason_for_churn	Pricing/Value, Network Coverage/Speed, Customer Service, Promotion Elsewhere, Billing Issues; empty if churn=No
competitor_switched_to	if churn=Yes, not equal to current provider; weighted to large MNOs
retention_offer_given	Yes/No; usually "Yes" for churners

Table 4 churn_decision_my_telco.csv

Column	Meaning / Description
customer_id	Unique identification number assigned to each customer.
Age	Age of the customer (in years).
Gender	Gender of the customer (Male or Female).
State	Malaysian state or region where the customer resides.
Provider	The telecom company the customer is subscribed to (e.g. CelcomDigi, Maxis, Unifi).
plan_type	Type of plan the customer is using (e.g. Prepaid, Postpaid, Family Plan).
contract_type	Type of service agreement — month-to-month or long-term (1-year, 2-year).
internet_service	Internet connectivity type (3G, 4G, or 5G).
tenure_months	Duration (in months) the customer has been with the provider.
monthly_fee_rm	The fixed monthly fee charged to the customer (in Malaysian Ringgit).
data_usage_gb	Average internet data consumed per month (in GB).
calls_made	Total number of phone calls made per month.
sms_sent	Total number of SMS messages sent per month.
bill_amount_rm	Actual monthly bill after usage (in RM)
payment_method	The mode of bill payment (e.g. credit card, online transfer, auto debit).
late_payment_count	The number of times a customer delayed bill payment.
customer_service_calls	The total number of times the customer contacted support.
complaint_count	Number of complaints filed by the customer.
network_quality_score	Customer's rating of network quality (scale of 1–10).
Churn	Indicates whether the customer has left the company (Yes/No).
satisfaction_score	Overall satisfaction score rated by the customer (1–10 scale).
reason_for_churn	If churned, the reason for leaving (e.g. poor service, high price).
competitor_switched_to	Name of competitor the customer switched to (if churned).
retention_offer_given	Whether the company tried to retain the customer with a special offer (Yes/No).

Table 5 Merged Dataset

The Telco Churn dataset consists of 5,000 customer records with various demographic, service, and behavioral features. The **average age** of customers is **33.95 years**, with the youngest at 18 and the oldest at 70. This suggests a customer base dominated by young adults and working professionals. The **average tenure** is **17.4 months**, showing that most users are relatively new or mid-term customers, while a few have stayed for over ten years.

In terms of cost, customers pay an **average monthly fee of RM82.56**, ranging from **RM18.27 to RM230.78**, reflecting a mix of prepaid and high-end postpaid plans. Data usage shows large variation, with a mean of **72.12 GB** but a maximum exceeding **1 TB (1050.61 GB)**, indicating some heavy internet users. Customers make about **60 calls** and send **15 SMS** per month on average, showing a stronger dependence on data than traditional calls or messages.

The **average bill amount (RM85.70)** is slightly higher than the monthly fee, possibly due to extra usage or service charges. On average, customers have **1.28 late payments** and **0.4 customer service calls**, while **complaint counts remain low at 0.36**, showing moderate engagement with service support.

Quality and satisfaction indicators show an average **network quality score of 4.05 (out of 5)** and a **satisfaction score of 7.19 (out of 10)**, suggesting generally positive but improvable customer experiences.

Categorical data reveals that most users are **female**, mainly from **Selangor**, and the leading provider is **CelcomDigi**, followed by other major competitors. Most of the users subscribe to **Prepaid** plans with **month-to-month contracts** and use **4G internet service**. The most common **payment method** is **online banking**, reflecting modern digital payment preferences.

The churn distribution shows that most customers **have not churned**, though about 29% have, often linked to network or pricing issues. Roughly **71%** of entries for **reason_for_churn** and **competitor_switched_to** are missing, indicating they apply only to churned users. Additionally, the **retention_offer_given** field indicates that most non-churned users did not receive special offers.

2.3 Target Variable of the Dataset

Churn is the target variable in this dataset since it is the primary outcome of interest that the study will predict: either a customer makes a decision to stay with the telecom provider or not. In telecommunications, the churn is an important business metric because loss of customers directly influences the revenue and the market share. In the data, yes churn means that the customer has already leaved the company, ceased the services and could have relocated to another service provider. No churn depicts that the customer is still using the service. Through the analysis of factors that determine a customer churn or not, like billing behaviour, network quality, complaints and plan type, we would have a clearer insight into the factors that contribute to the loss of a customer and devise strategies to help mitigate the churn in the Malaysian telecom market.

2.4 Exploratory Data Analysis (EDA)

A detailed Exploratory Data Analysis (EDA) was carried out before combining the three separate main tables into one modelling dataset, to learn the nature of each table. The relational integrity was verified and guarantee that the synthetic data is a true representation of actual Malaysian telecom trends. The purpose of this stage is to ensure that the quality of data remains intact, brings to light critical behavioral trends, and warrants the merging process necessary to develop predictive models that are trustworthy.

2.4.1 EDA on Customer Master Table

The Customer Master table includes the demographic and subscription level data like age, gender, state, service provider, plan type, contract type, internet service, tenure and monthly fees.

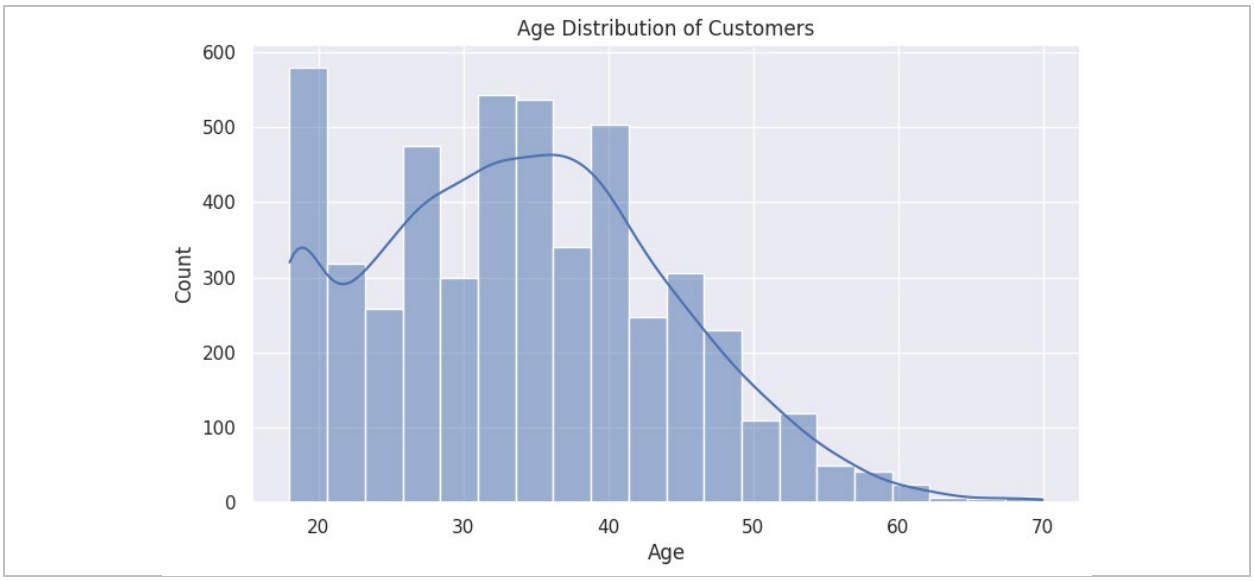


Figure 1 Age Distribution

The histogram indicates that the age of customers is clustered around 20-40 years old, with almost no older users. This distribution is consistent with the real Malaysian telecom demographics, where young adults and working professionals tend to have mobile subscriptions.

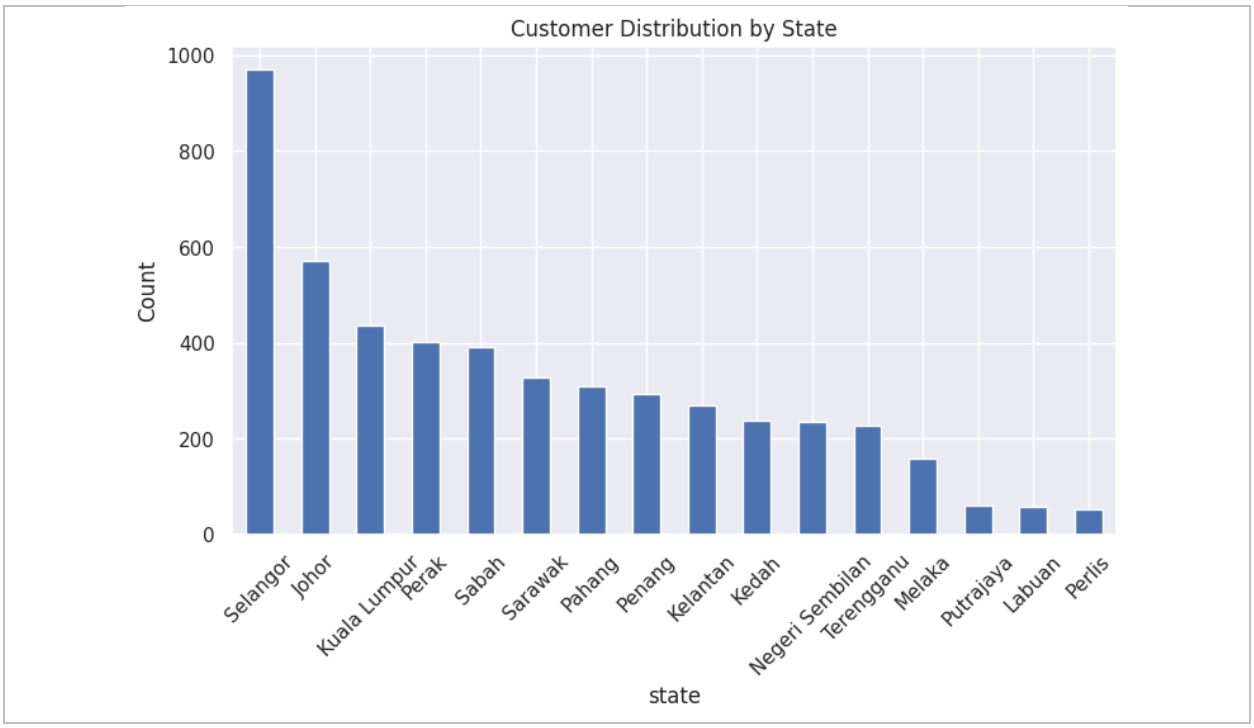


Figure 2 State Distribution

The majority of entries are comprised of Selangor, Johor, and Kuala Lumpur. This trend is similar to the official DOSM distribution of population where these states are more urbanised and populated. Therefore, synthetic dataset represents actual geographical clusters of demand.

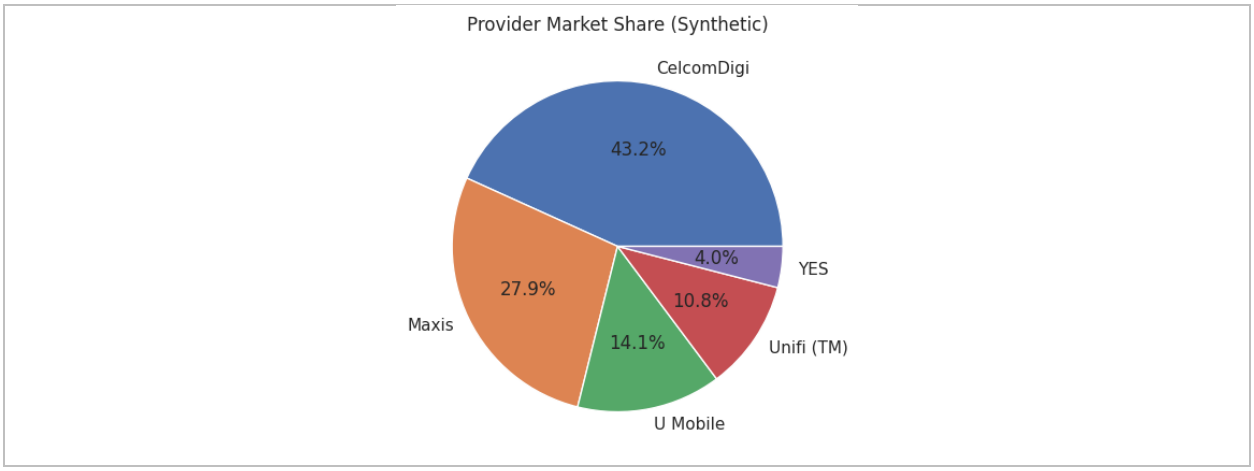


Figure 3 Service Provider Distribution

The majority is CelcomDigi with a share of Maxis, U Mobile, Unifi (TM) and YES. The ranking is similar to the actual Malaysian telecom market, where CelcomDigi and Maxis are the leading MNOs.

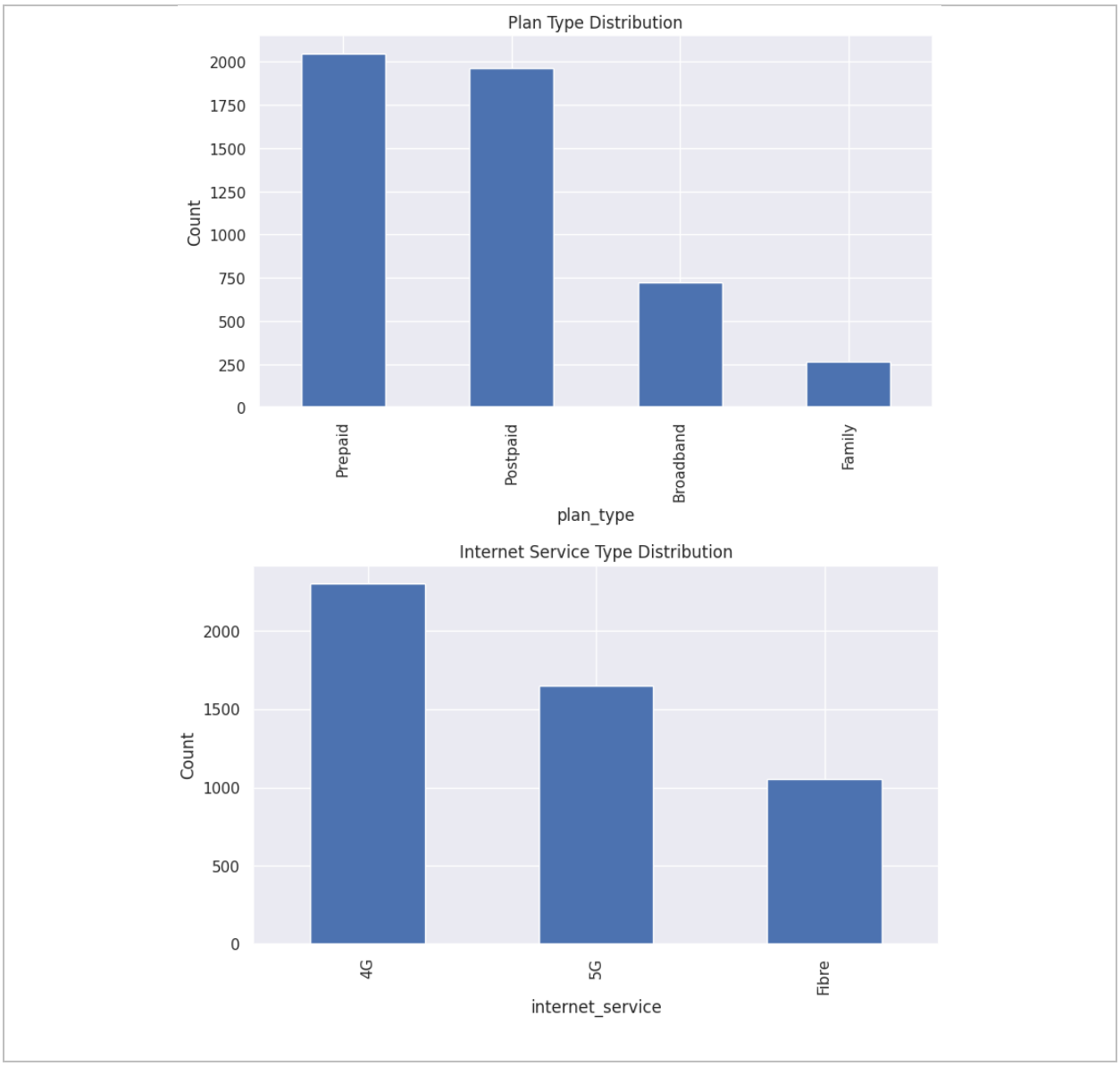


Figure 4 Plan Type and Internet Service

The dataset is dominated by prepaid and Postpaid plans with smaller segments of Broadband and Family plans. There are realistic tendencies in internet service types (4G, 5G, Fibre): 4G is the most widespread, 5G is more concentrated in urban states, and Fibre is typically related to Broadband/Family plans.

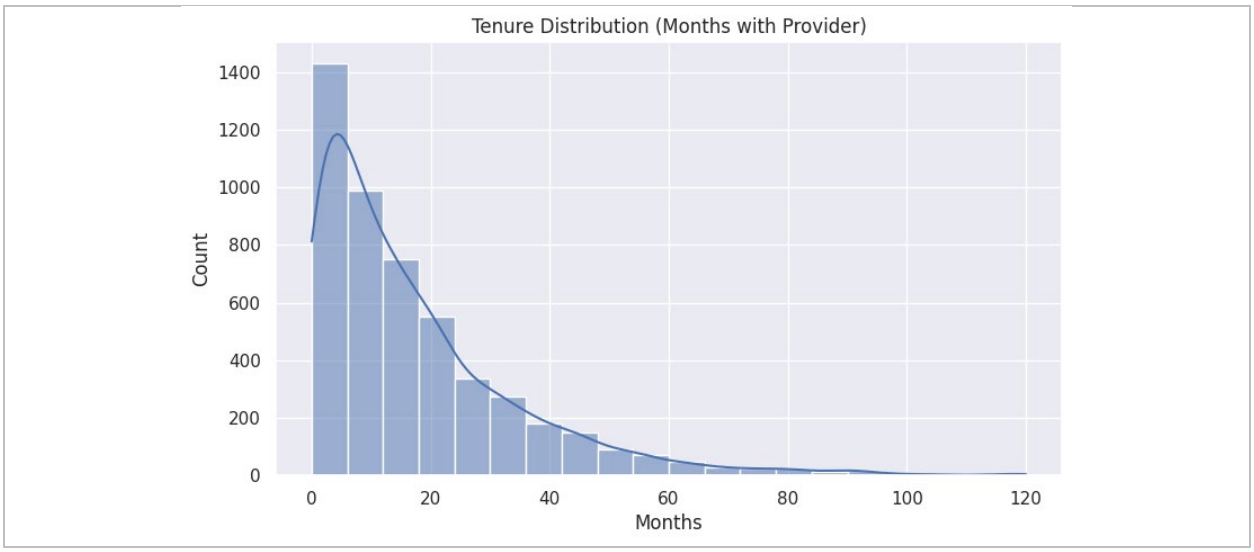


Figure 5 Tenure Distribution

The distribution of tenure is skewed on the right, with a large number of customers being new (<12 months), and fewer having a long tenure (>36 months).

Overall, demographic and subscription-based variables show expected behavior in a real Malaysian telecom dataset supporting the realism of the synthetic data.

2.4.2 EDA on Usage Transaction Table

This table summarises behavioural indicators like data usage, calls, SMS, payment method, complaints, late payments, and network quality.

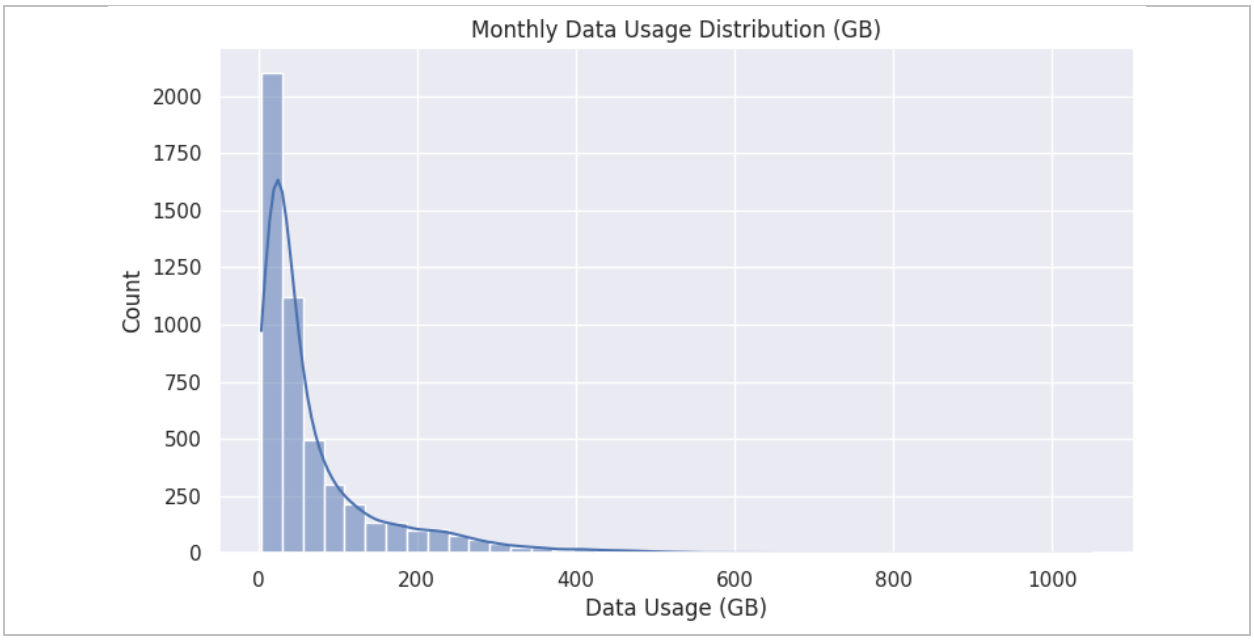


Figure 6 Data Usage

Data usage (GB) is skewed to the right. The usage of most users is 10-150 GB with a long tail of heavy users (e.g., Fibre or unlimited 5G subscribers). This trend closely correlates with the current trend of telecom where data is the most dominant service.

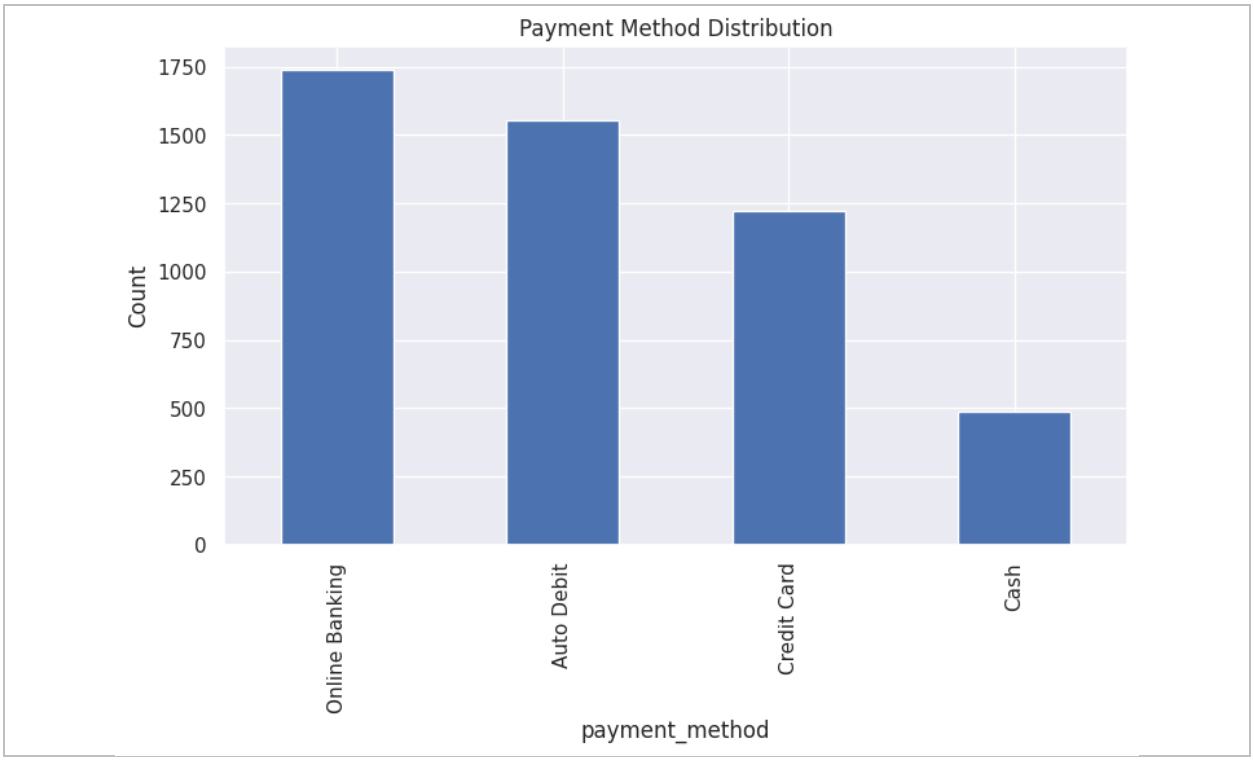


Figure 7 Payment Method

The most popular payment methods are Online Banking and Auto Debit and least popular Cash. This aligns with the trend of Malaysia embracing digital payment.

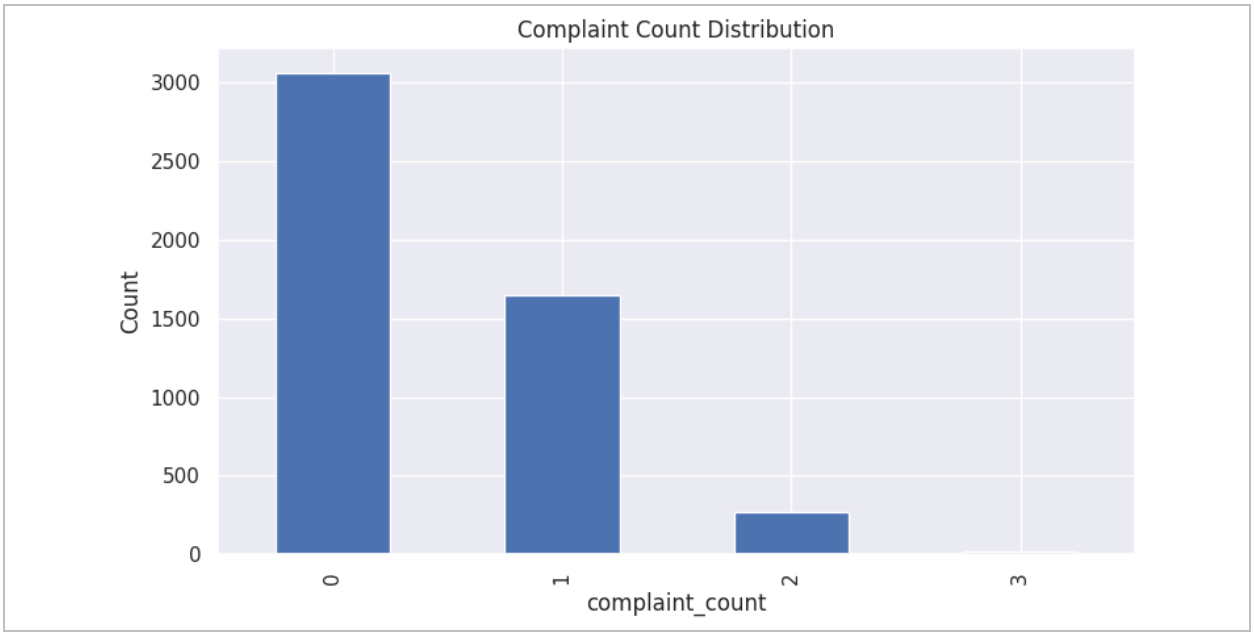


Figure 8 Complaints

The majority of customers complain 0-1 times, and smaller group complains 2-3 times. This automatically isolates those who are satisfied and those who are not.

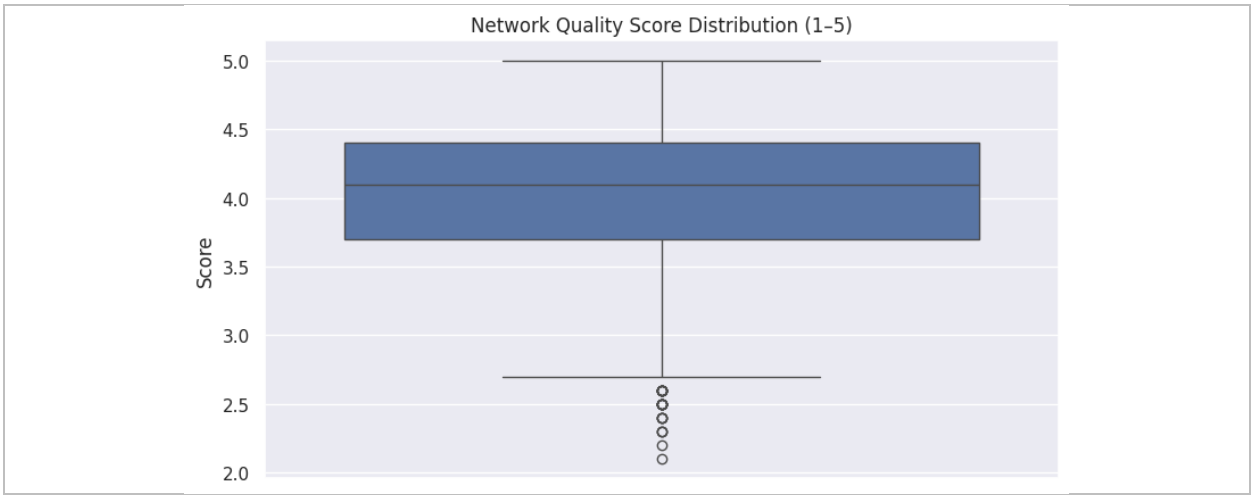


Figure 9 Network Quality Score

Most users have a range between 3.0 and 4.5 meaning fairly good service. A very small group of low scores (<3.0) will tend to be found in churned groups.

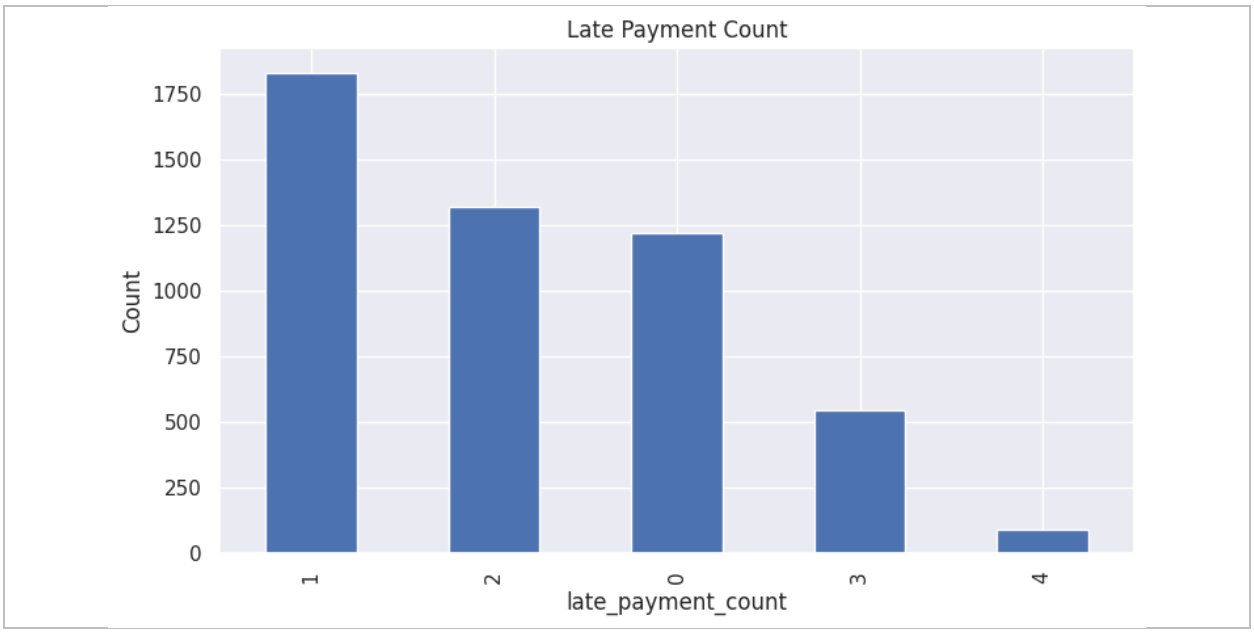


Figure 10 Late Payment Count

The distribution indicates that there are numerous clients with zero or one late payment, and others with multiple late payments, in line with different financial behavior. The churn risk is characterized by late payments.

2.4.3 EDA on Churn Decision Table

The Churn Decision table includes churn labels, satisfaction scores, churn reasons, competitor choice, and retention offer.

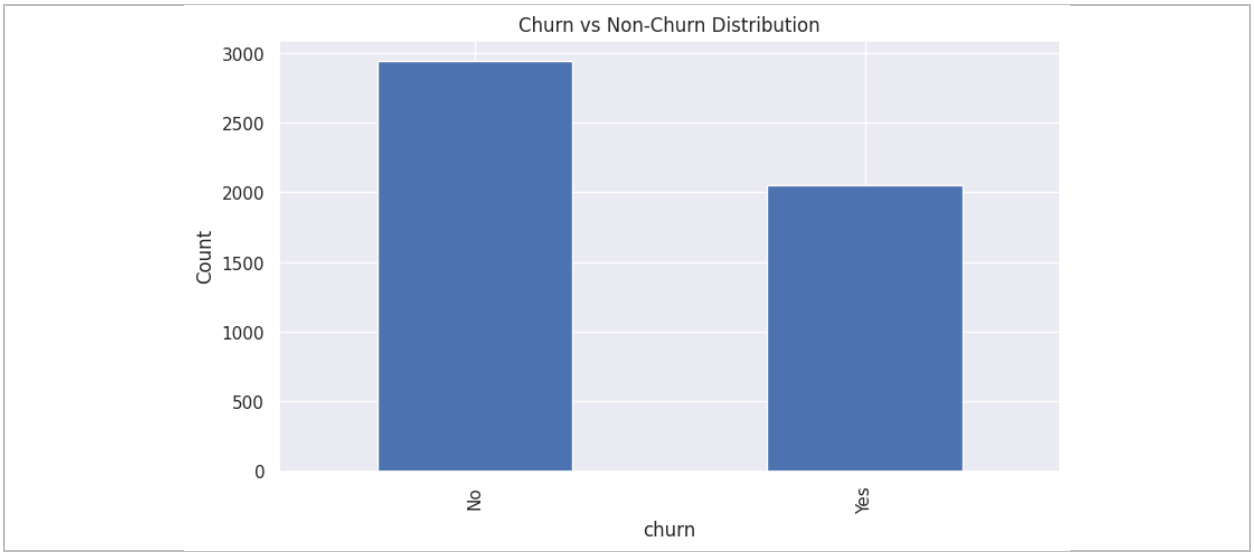


Figure 11 Churn Distribution

The churned rate of about 29% is in line with average industry churn rates (20-35%). This will prevent the probability of having an overly imbalanced target variable to make the modelling phase meaningful.

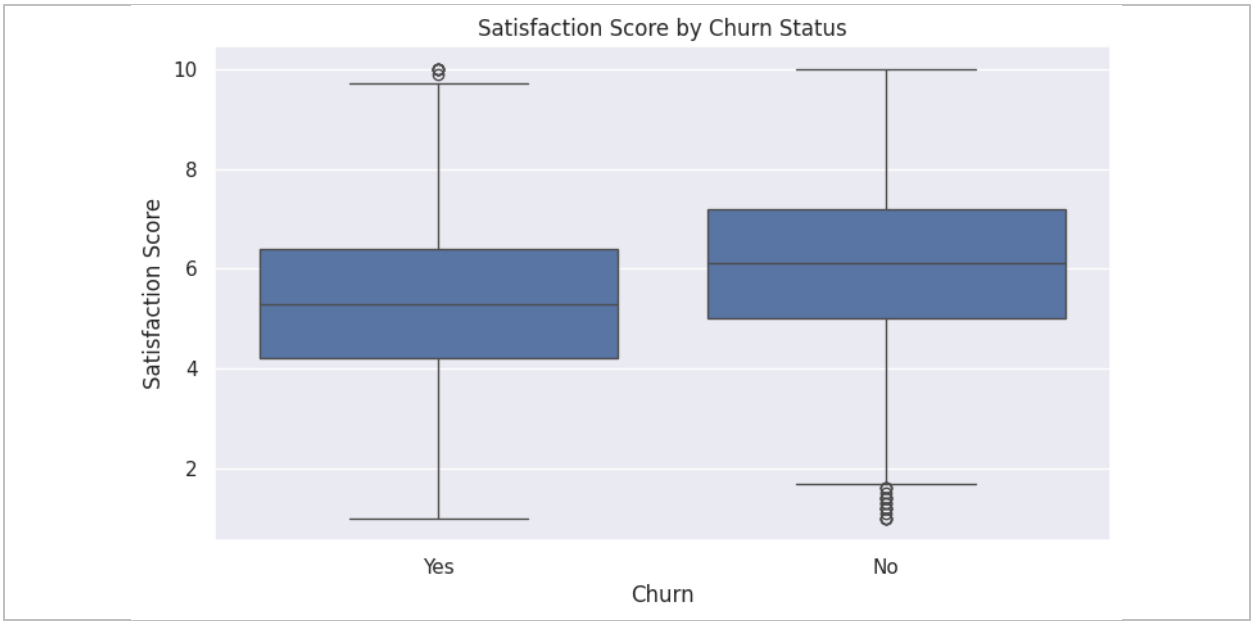


Figure 12 Satisfaction Score

There is a strong distinction between churners and non-churners: churned customers are much less satisfied. This confirms that the score of satisfaction is an excellent predictor of the churn.

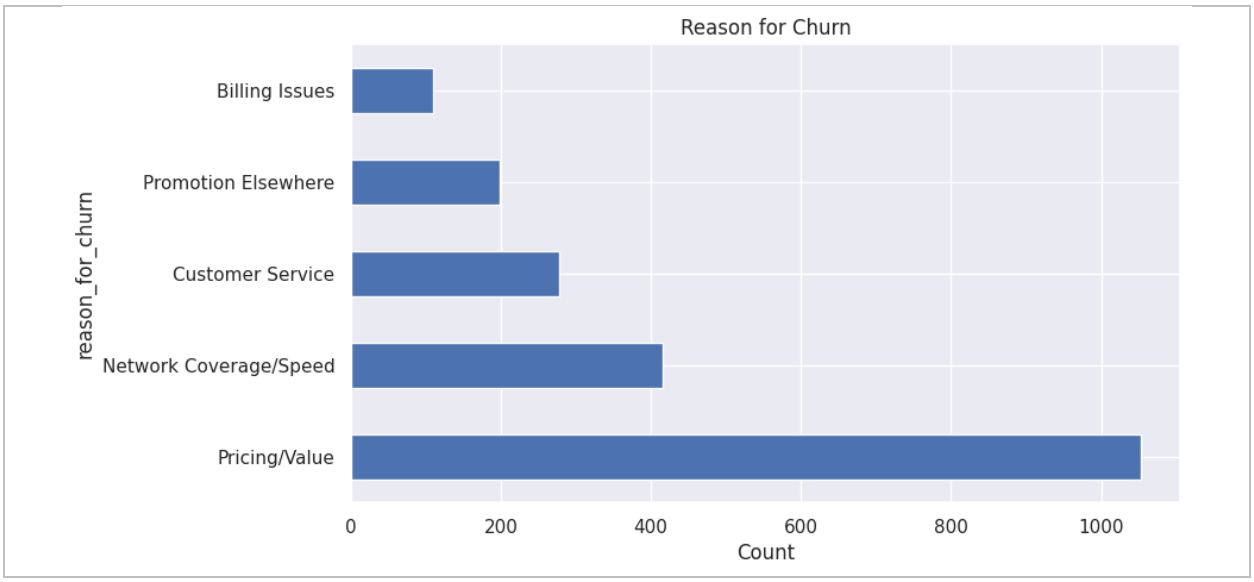


Figure 13 Reason for Churn

The reason might be value dissatisfaction, prices, network cover, or the speed.

For further details, please refer to the notebook telco_malaysia_EDA_ipynb.

Chapter 3: Implementation and Evaluation

3.1 Processing Workflow of Orange Tool

The workflow begins with the File widget, where is use to load the dataset. Then, the data is sent to several components at the same time. First, it goes to the Data Table so the raw dataset can be viewed . The data is also go to three learners: Decision Tree, Random Forest, and Logistic Regression. These learners create different predictive models for the target variable Churn. The Decision Tree model is additionally connected to the Tree Viewer, which allows visualisation of the tree structure. All three trained models are then fed into the Test and Score widget, where their performance is evaluated using metrics such as accuracy, AUC, precision, and recall. The evaluation results are sent to two analysis tools: the Confusion Matrix, which shows how well each model correctly predicts "Churn: Yes/No," and the ROC Analysis, which visualises model performance across different thresholds. Finally, the models are connected to the Predictions widget, which applies the chosen model to produce actual churn predictions for the dataset. Besides, this workflow allows comparison of multiple algorithms and provides a clear assessment of the best model for predicting telecom customer churn.

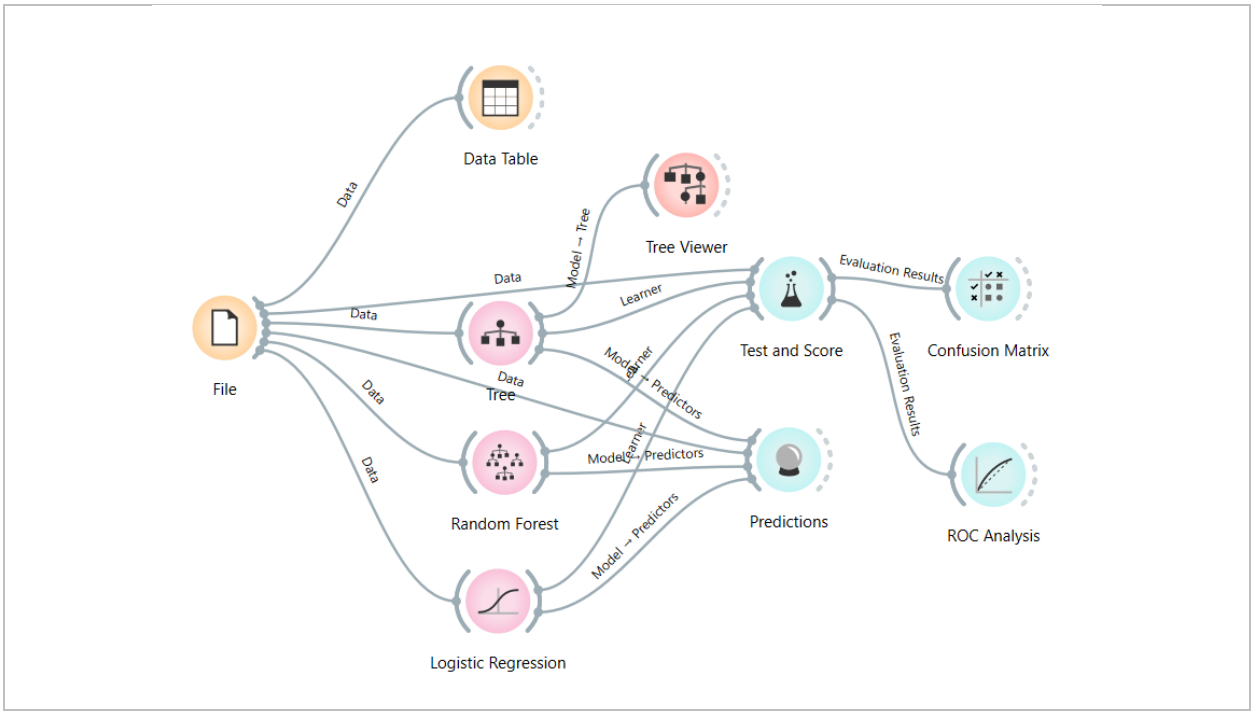


Figure 14 Overflow of the Process

3.2 Models Implementations

In this project, we apply three machine-learning models, which are Decision Tree, Random Forest, and Logistic Regression, to make predictions on whether a customer will churn or not. Each of the models is a variant of learning patterns on the data. The Decision Tree model operates by repeatedly dividing the data into branches depending on the most significant features to form an easy to understand flow of decisions resulting in a churn or non-churn outcome. Random Forest model is an expansion of decision tree model; it does not just construct a single tree instead of that, it constructs a number of trees and sums the results of each tree to come up with a more consistent and accurate prediction. This aids in minimizing overfitting and also more intricate patterns in the data are captured. Meanwhile, the Logistic Regression is a statistical model which approximates the churn probability based on the effect of each feature on the event of customer departure. It is extensively applied in binary classification problem since it is easy, rapid and works well when dealing with high volume of data. When these three models are applied, we can compare various learning strategies and determine which of them is the most appropriate to predict telecom customer churn.

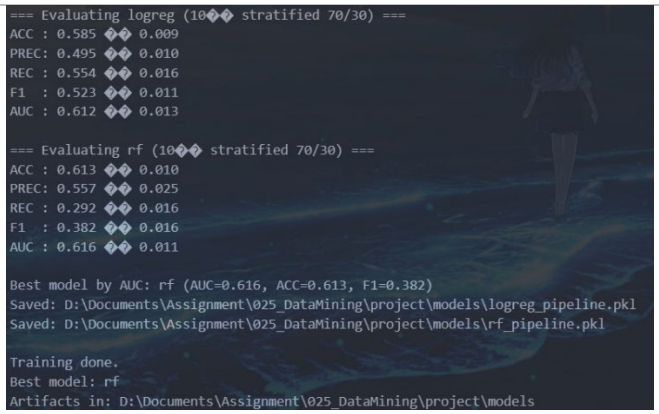
3.3 Evaluation Metrics of Results

According to Python and Orange model evaluation results, there are certain differences, despite the fact that all tools evaluate the performance of the same models; the Logistic Regression, the Random Forest and the Decision Tree. These variations are primarily due to the fact that the two environments employ dissimilar assessment environments. In Python, they are run on a 70/30 train-test split repeated 10 times, whereas in Orange, they are run on 5-fold cross-validation or repeated random sampling based on the configuration. The use of a train-test split may introduce randomness and rather unstable results, particularly with a dataset of imbalanced classes. Cross-validation, however, tends to make more consistent and reliable estimates since it is trained and tested on several folds of data. Also, Python only evaluates the pipelines of the desired models, as opposed to Orange that evaluates all models with the same controlled cross-validation scheme constructed into the interface, which may give more uniform comparisons.

In comparing the two platforms, the Orange results are in most cases more appropriate to use in reporting in a data mining project, as cross-validation is the most common way to compare

numerous models fairly. It eliminates bias when using a single train-test split, and yields more realistic representation of generalisation in practice.

In both Python and Orange, Logistic Regression has always done better. It has the largest AUC (0.630), accuracy, F1-score, precision, and recall in the Orange results, which shows that it describes the churn patterns more accurately than Decision Tree and Random Forest. The reason why the Logistic Regression works is that churn prediction is a binary issue where most relationships among the features are linear in nature (tenure, complaints, late payment, network quality etc.). The model is effective in the treatment of these continuous variables, does not overfit, and makes consistent predictions at repeated estimation. This renders Logistic Regression a very useful reduction of churn analysis particularly when the interest is to control the significant predictors as well as to generate results that are readable to business decision-making.



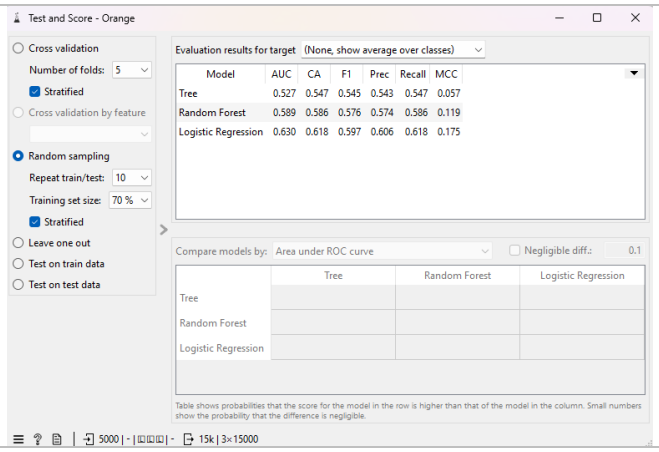
```
=== Evaluating logreg (10►► stratified 70/30) ===
ACC : 0.585 ►► 0.009
PREC: 0.495 ►► 0.010
REC : 0.554 ►► 0.016
F1  : 0.523 ►► 0.011
AUC : 0.612 ►► 0.013

=== Evaluating rf (10►► stratified 70/30) ===
ACC : 0.613 ►► 0.010
PREC: 0.557 ►► 0.025
REC : 0.292 ►► 0.016
F1  : 0.382 ►► 0.016
AUC : 0.616 ►► 0.011

Best model by AUC: rf (AUC=0.616, ACC=0.613, F1=0.382)
Saved: D:\Documents\Assignment\025_DataMining\project\models\logreg_pipeline.pkl
Saved: D:\Documents\Assignment\025_DataMining\project\models\rf_pipeline.pkl

Training done.
Best model: rf
Artifacts in: D:\Documents\Assignment\025_DataMining\project\models
```

Figure 15 Python Result



Test and Score - Orange

Cross validation
Number of folds: 5
Stratified
Cross validation by feature
Random sampling
Repeat train/test: 10
Training set size: 70 %
Stratified
Leave one out
Test on train data
Test on test data

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Tree	0.527	0.547	0.545	0.543	0.547	0.057
Random Forest	0.589	0.586	0.576	0.574	0.586	0.119
Logistic Regression	0.630	0.618	0.597	0.606	0.618	0.175

Compare models by: Area under ROC curve Negligible diff.: 0.1

	Tree	Random Forest	Logistic Regression
Tree			
Random Forest			
Logistic Regression			

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

5000 | - | 15k | 3×15000

Figure 16 Orange Result

Chapter 4: Performance Analysis and Discussion

4.1 Suggestion after Data Mining

In this project, three suggest data mining models to use in finding the rules and predicting the churn include Decision Tree, Random Forest, and Logistic Regression. Decision Tree model is helpful in rule mining since it inherently produces human-readable if-else rules indicating the relationship between features and churn outcomes. Its advantage is that it could be easily interpreted and one could know the conditions (e.g., high complaints or low tenure) that trigger churn. The Decision Trees are extremely sensitive to noise as well as may overfit and cause unstable performance. Next, the Random Forest model can be considered as a better version of this by using a lot of decision trees to produce more generalised and reliable predictions. It does not overfit and manages complex relationships, and thus it is better in accuracy and AUC. The principal weakness of it is that the rules have become less interpretable due to hundreds of trees that are incorporated in the ultimate prediction and thus it is not easy to obtain clear sets of rules. Last, the Logistic Regression can be used as a powerful baseline to work with binary classification problems such as churn prediction. It generates probability results and gives definite details as to the impact of each feature on churn. The Logistic Regression is suited well to structured tabular data and can withstand noisy inputs. Its weakness lies in the fact that it assumes that relationship is linear, as such, it might ignore any complex interactions that can be observed in tree-based models. For conclusion, Random Forest offers the most impressive predictive accuracy, Decision Trees are better at extracting rules and Logistic Regression is easier and more understandable.

4.2 Conclusion

In conclusion, the project was able to use the data mining methods to analyse and predict customer churn in the telecom sector, in the Malaysian context, utilising the realistic synthetic data, backed up with DOSM demographic and telecommunications data. The study examined various predictive methods and contrasted their performance by using three models namely, Decision Tree, Random Forest, and Logistic Regression. The outcomes indicate that the logistic regression offers the best stable and reliable estimates, random forest is efficient in the ability to describe complicated patterns and decision trees are useful in deriving interpretable rules. The analysis indicates that the main contributors to churn include low tenure, network poor quality, high rate of complaints and

delayed payments. These results prove that data mining can help telecom companies to identify at-risk customers and provide better retention strategies. Overall, the project demonstrates that predictive modelling is a useful instrument in customer behaviour understanding, and it will offer a starting point in the subsequent improvement.

Chapter 5: System Design and Prototype

The main goal of this system is to create a decision support system that can make use of existing built datasets and predictive models that were previously trained to predict and visualize the probability of customer churn in the telecommunications sector of Malaysia. Particularly, the system is expected to achieve a merging of data mining models into a working application in which users provide customer demographic and usage data to receive the results of churn prediction. The system delivers interpretable results, such as the predicted probability of churn, churn labels, and suggested levels of retention of the customer to aid in business decision-making.

5.1 Scope of System Design

This system is restricted to the implementation and utilization of basic data mining procedures. It is not a detailed or in-depth study of real-life telecom customer churn data in the sector. At this point, the system will just be a desktop prototype that will prove its useability under actual operational settings. It is centered on demonstrating how results of data mining may be converted into usable software tools. It is specifically used in the prediction of customer churn and customer retention in the telecom industry of Malaysia.

5.2 System Architecture Overview

The system is written in a modular architecture which results in the system being easily scaled, maintainable, and easy to use. It is a combination of three layers:

a) Data Layer: This contains the raw data of the customers such as demographic data, usage statistics and churn data in structured CSV format.

b) Modeling Layer: This is the major system component that contains the trained predictive models (Logistic Regression and Random Forest). These models are used to make predictions based on input data.

c) Application Layer: User interface (UI) that will allow connecting to the model and make churning predictions.

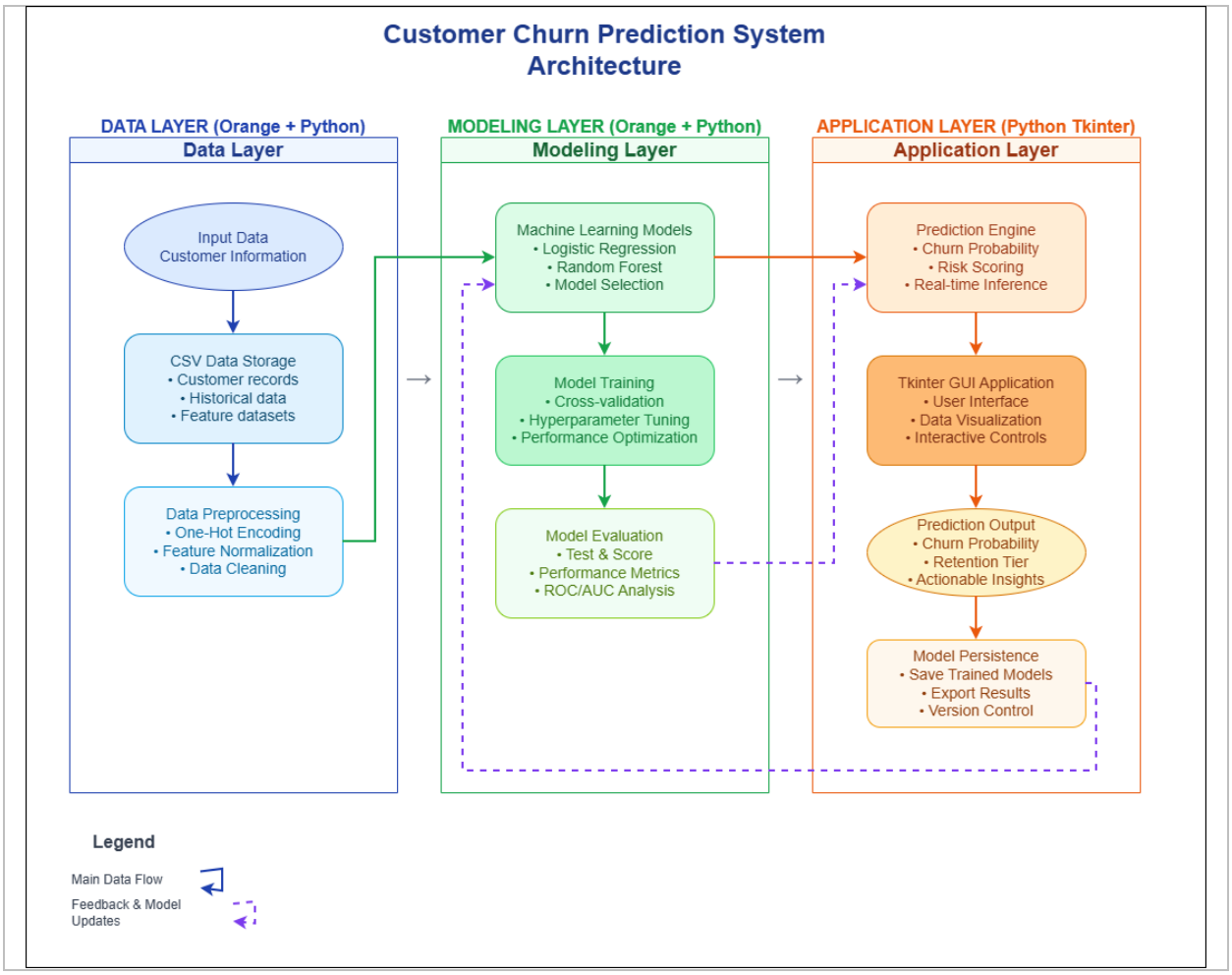


Figure 17 System Architecture Flowchart Diagram

5.3 System Components

The Telecom Service Provider Churn Prediction System comprises three key elements, Data Layer, the Modeling Layer and the Application Layer (as illustrated in figure 4). Each of the components has been described below.

a) Data Layer

Data Layer is the layer that holds the raw data and provides their structure in a form of organization.

Customer Master Table: This table is a list of customer demographics (age, gender, state, provider, and plan type).

Usage Transaction Table: This table contains the data about the usage patterns of the customers, such as data consumption, the number of calls and SMS, and the payment history.

Churn Decision Table: Records customer churning information (Yes/No); other details, including satisfaction score, the reason why they churned, and the competitor the customer moved to.

These data are first saved in the CSV form and then combined into one dataset (telcochurnmymerged.csv) to be used in model training and prediction.

b) Modeling Layer

The main analysis part of the system is the Modeling Layer. It contains the machine learning models that make the churn predictions using the customer data. The models implemented and evaluated are the following:

1. Logistic Regression (LogReg): It is a statistical model that estimates the likelihood of churning, by relating the features to the target variable. Logistic Regression is suitable to the linearly separable data and also gives the probability scores and interpretation coefficients.
2. Random Forest (RF): This is an ensemble learning algorithm that creates a batch of decision trees that aids in making stronger predictions. The Random Forest model deals with irregular and non-linear relationships in the data and less prone to overfitting than single decision trees.

These models are implemented using scikit-learn in Python, and the models are saved to be used later in the application layer using joblib in the form of a.pkl file.

c) Application Layer

Application Layer is the interface (UI) that enables one to interact with the trained models. The UI is developed on the Tkinter Python library that is designed to develop a graphical user interface. The main characteristics of the application layer are:

1. Insert Data: The GUI provides interface input fields where users can input the customer demographic and usage data.
2. Model Inference: After the user input data, the system employs the trained model (Logistic Regression or Random Forest) to forecast the probability of churn and categorize the customer to be either churning (Yes/No).
3. Output Prediction: The churn prediction and the probability score and retention tier recommendation (e.g., High, Medium, Low) is shown in the system.

d) Scripts

The Python scripts that were developed to support the functionality of the system are as follows:

1. datasetGenerationCodeV2.py: The script is used to create a synthetic dataset that is used to train and test. It generates customer demographic, usage statistics and churn labels by using some risk factors like type of contract, mode of payment and quality of service.
2. modeltraining.py: This script is used to train the machine learning models (Logistic Regression and random Forest) with the merged dataset. It executes the repeated cross-validation procedure, measures model performance on several metrics (AUC, Precision, Recall, F1-Score), and stores the trained models in the form of.pkl files to use in deployment.
3. telcochurngui.py: This program is the primary Tkinter GUI, which interacts with a user, giving them the opportunity to provide data either manually or through batch upload. The script loads the models which were trained, makes predictions and shows the results. It also gives an opportunity to export prediction results to further analysis.

5.4 Key Features

The Telecom Service Provider Churn Prediction System is aimed at delivering a holistic solution in the process of predicting customer churn in the Malaysian telecommunications sector. The system combines information mining activities and machine learning models to provide actionable suggestions to telecom firms that seek to minimize churn and enhance retention strategies.

The heart of the system is a data preprocessing pipeline that guarantees that the data is cleaned, transformed and prepared to be model trained. Missing values are treated by imputation, in which values of categories are imputed with the most common, and numerical values with the median. Numerical representations are made by one-hot encoding categorical features such as gender, state, provider and plantype. Further, numerical variables, like age, tenuremonths, and monthlyfeerm are standardized with Z-score normalization, which is adapted to ensure no feature has unproportional contribution to the model because of its magnitude.

Two machine learning models are used to predict churn in the system: Logistic Regression and Random Forest. Logistic Regression can be used in this task especially due to its capability to formulate the relationship between customer characteristics and churn as a straightforward probability that can be interpreted and can be easily applied to problems in binary classification. The Random Forest model on the other hand is a composite of decision trees that can be used to identify more complex and non-linear trends in the data and is therefore more effective in cases where the relationship between the features is non-linear. They are evaluated by several measures such as accuracy, precision, recall, F1-score, and AUC, to make sure that the model performance is comprehensively tested before its implementation.

The models are trained and then applied to determine the probability of a customer churning based on their usage and demographic statistics. The system offers a churn probability, a binary churn label (whether the customer will churn or not), and a retention tier recommendation to each customer. This retention level is a classification that enables the telecom company prioritize activities: an example of this is high-risk customers who are at risk of churn and therefore potentially targeted with special retention offers whereas low-risk customers can be passively monitored.

Its user interface (UI) is constructed with the Tkinter library that gives the user a user-friendly interface to add customer information in a batch or individually. The interface enables the individual customer record predictions of churn and predictions of churn at scale using the entire dataset uploaded in CSV format. The results are presented in clear formats, including the churn prediction, and probability of the customers, as well as retention tier.

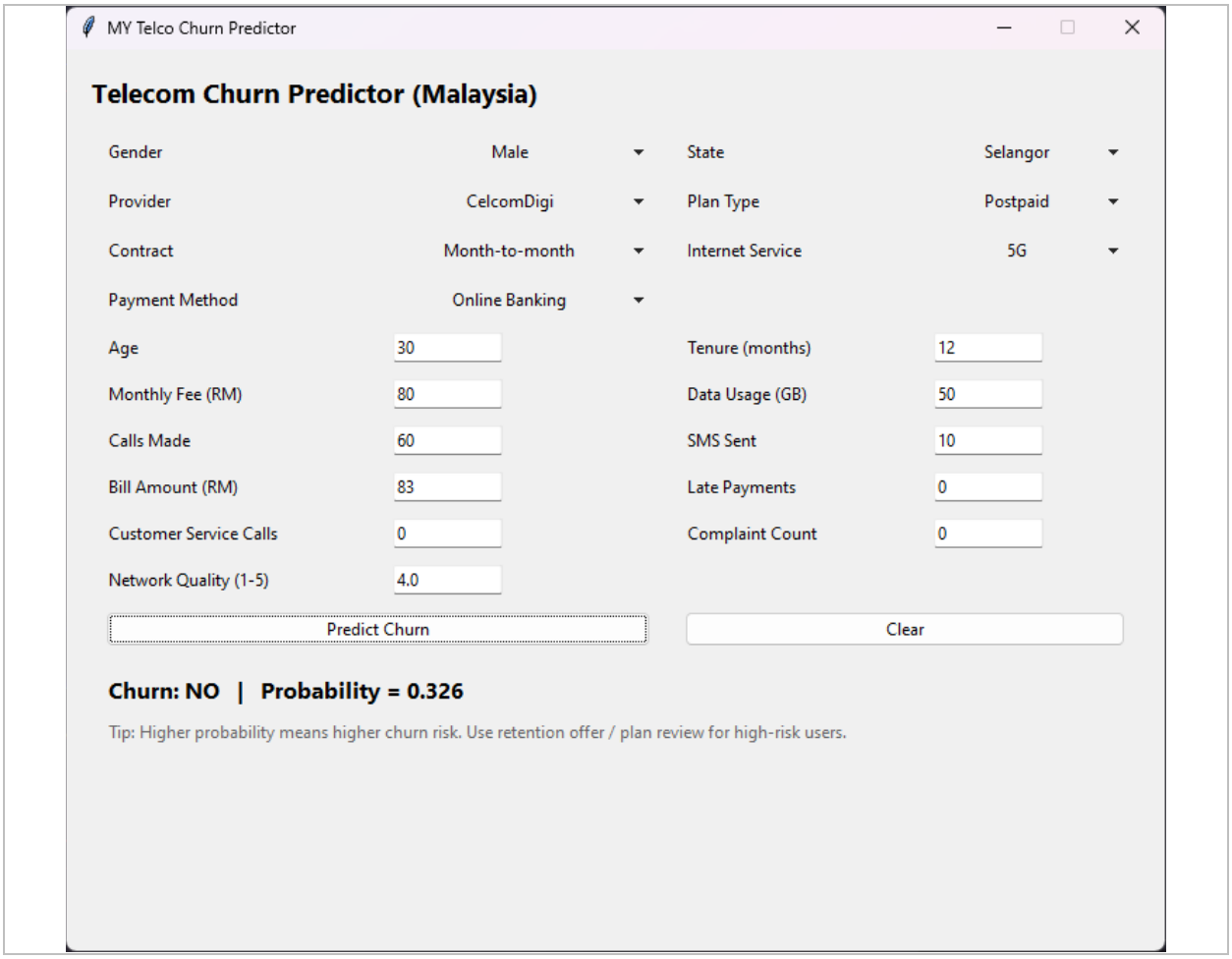


Figure 18 UI of System

5.5 Conclusion

Telecom Service Provider Churn Prediction System is an efficient system that combines the data mining technique and machine learning models to deliver a successful tool in prediction and analysis of churn within the telecommunications industry of Malaysia. The system, which includes data generated artificially and real-world population distributions as well as industry-specific information, can provide a useful option in learning about customer behavior and enhancing retention strategies.

The system can make churn predictions with varied accuracy using the Logistic Regression and the Random Forest, thus leaving businesses with helpful measures such as the probability of churn and levels of retention tier classifications. These insights will enable the telecom companies to focus on those customers who are at risk and offer them individualized retention packages which will probably decrease the churn rate and customer loyalty will increase.

It is also important to note that the system emphasizes data preprocessing and model evaluation, which makes sure that models employed are strong and stable. Although the Logistic Regression model was superior in this specific context, the Random Forest model is more flexible in dealing with intricate data associations, and is worth considering as an alternative in the future based on the characteristics of subsequent datasets.

However, the system has its limitations despite its success. It is largely based on simulated data and might not represent the real-life telecom customer behavior quite accurately. Also, the system is limited in terms of scalability due to its desktop prototype system, which may not be scalable to a large enterprise system. However, this project proves that predictive modeling has a tremendous potential in the telecom sector as it gives practical insights that can be used to better customer retention and enable data-driven decision-making.

Chapter 6: References

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